SecureDT - Review 0 Presentation Guide

Live demo walk-through for reviewers

Suggested walk-through for a Review 0 demo. Start the app, then work through each capability below so reviewers can verify every claim live.

1. Dashboard

Open / - shows current security risk, online MEC nodes, GA progress, attack counters, live event feed, topology and resource cards. Auto-refreshes via /api/state polling.

2. MEC Infrastructure

Open /mec-infrastructure. Show the MEC registry, pairing-code registration, heartbeat-based ONLINE/OFFLINE status and token rotation settings (HEARTBEAT_INTERVAL_SECONDS=10, HEARTBEAT_TIMEOUT_SECONDS=30, MAX_TOKEN_AGE_SECONDS=3600).

3. Digital Twin & Attack Response

Open /digital-twin and /attack-simulation. Launch a simulated attack (DDoS, Eavesdropping, Tampering, Data_Poisoning, Credential_Abuse) on a node, watch risk climb, quarantine the node, stop the attack and watch risk decay.

4. Genetic Algorithm Scheduling

- Open /genetic-algorithm; configure population (40), generations (50), mutation (0.12), crossover (0.85).
- Run the proposed security-aware GA and show best-fitness decreasing per generation.
- Open /task-routing to see placement and the OFFLINE-never-assignable rule.

5. AI Intelligence

- Open /ai-intelligence and analyze a task -> risk level (safe/trained/critical) + confidence + feature reasons.
- Show models/risk_metrics.json and the deterministic rule fallback when the .pkl is absent.
- Optional: with GROQ_API_KEY + AI_PROVIDER=groq, LLM insight appears on the page.

6. Datasets, Analytics & Experiments

- /datasets -> registry, big-data generation (up to 10M), quality, normalization.
- /analytics -> compare proposed_ga vs random / greedy / privacy_aware.
- /experiments-run documented in EXPERIMENT_GUIDE.pdf (campaigns append to experiment_results_2000.csv).

7. Security Controls

Explain the enforced protections: Django CSRF, session + token auth on /api, headers set by Django middleware, upload validation (CSV only, max 8 MB / 100,000 rows), signed sessions, and audit logging to datasets/audit_log.csv.

Key numbers to state

	Value
	latency 0.25, energy 0.15, privacy 0.20, security 0.30, deadline 0.
	tournament(3), crossover 0.85, mutation 0.12, elitism 4
	10 s / 30 s
	enabled, max age 3600 s
	LOW/MEDIUM/HIGH/CRITICAL at 0.00/0.30/0.60/0.80
	8 MB, 100,000 rows
	PyInstaller -> dist/MEC_Agent.exe